Exam craft for the SMC and AMC 12

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L3-ST1 Level 3 Senior · Strategy

How the SMC and AMC 12 are scored, when to guess, and the techniques (specialising, small cases, working backwards, estimating, eliminating) that save time and marks.
By the end you can decide when guessing helps under the SMC and AMC 12 scoring rules; solve questions by specialising, testing options and estimating; manage your time and spot questions that are time sinks.
Prerequisites: Factorising, substitution and systems (L2-AL1), Sequences, sums and averages (L2-AL2), Counting without missing anything (L2-CO1). **Contests:** UKMT SMC, Senior Kangaroo, AMC 12, AMC 10. **Time:** about 45 minutes.

1 Lesson

Consider the following question, which is typical of the SMC.

The mean of five numbers is m. Two of the numbers have mean $m+3$. What is the mean of the other three?
(A) $m-3$ *(B)* $m-2$ *(C)* $m-1$ *(D)* $m+2$ *(E)* $m-\frac{6}{5}$

You could set up totals: the three numbers sum to $5m-2(m+3)=3m-6$, so their mean is $m-2$. That works, but note that the options are fixed expressions in m, so the correct one must be right for every m (specialising). Put $m=0$. The five numbers sum to 0, the two sum to 6, so the three sum to -6 and have mean -2. Only (B) gives -2 at $m=0$. This takes a few seconds and needs no algebra.

This unit covers such habits for multiple-choice papers: knowing the scoring, managing time, specialising, testing options, estimating, and deciding when to leave a question. None of this replaces mathematics, but it stops you losing marks you have earned.

1.1 Know the rules before you sit down

The UKMT Senior Mathematical Challenge. You have 90 minutes for 25 questions, with no calculators, no squared paper and no measuring instruments. According to UKMT's SMC awards page and the instructions printed on the 2026 paper, everyone starts on 25 marks, a correct answer earns 4, a wrong answer loses 1 and a blank scores 0, so the maximum is 125. Questions 1 to 22 have five options, A to E. On the 2026 paper Questions 23 to 25 have no options, and you mark a three-digit answer from 000 to 999. Formats do change, so read the instructions on your own paper before the start. The questions are roughly in order of difficulty. The mark needed to qualify for BMO Round 1 is set afresh each year once the results are in, so there is no fixed target to play safe against.

It is worth doing the arithmetic in advance. Starting on 25, reaching 101 means $4c-w\geq 76$, where c counts correct answers and w wrong ones. Nineteen correct with no errors is enough; twenty correct allows four errors. Hence a BMO1 qualifier needs about twenty careful answers,

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not an attempt at all twenty-five.

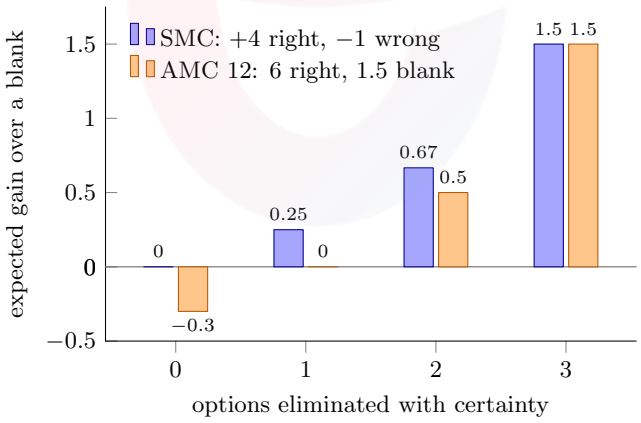
The MAA AMC 12. You have 75 minutes for 25 questions, each with five options. The instructions on the paper (reproduced on the AoPS wiki's AMC 12 page) give 6 points for a correct answer, 1.5 for a blank and 0 for a wrong answer, so the maximum is 150, which is the perfect score on MAA's statistics page. MAA's AMC 10/12 Teacher's Manual lists the only permitted aids as writing utensils, blank scratch paper, rulers, compasses and erasers: no calculators. According to MAA's competition policies, at least the top 5% of AMC 12 scorers are invited to the AIME.

1.2 When should you guess?

The guessing rule.
Suppose you have eliminated k of the five options with certainty and you guess at random among the remaining $5-k$.
On the SMC the expected score of the guess is

$$\frac{1}{5-k}\cdot 4-\frac{4-k}{5-k}\cdot 1=\frac{k}{5-k}.$$

This is 0 when $k=0$ and positive for every $k\geq 1$. A blind guess is neutral; a guess after one justified elimination gains on average.
On the AMC 12 a guess among $5-k$ options is worth $\frac{6}{5-k}$ on average, against a certain 1.5 for a blank. Since $\frac{6}{5-k}>1.5$ exactly when $5-k<4$, guess only once you have eliminated at least two options. With one eliminated it is break-even; with none, the guess costs 0.3 on average.



The SMC integer questions are different. There are 1000 possible answers, so a blind guess is worth about $\frac{4}{1000}-\frac{999}{1000}$, which is nearly a full mark lost. Fill those in only if you have actually worked something out.

Note that "eliminated with certainty" means you have a reason, not a feeling. Apply the rule and move on.

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1.3 Managing time

The SMC gives you 3.6 minutes per question on average and the AMC 12 three. The early questions should take about a minute each, which leaves time for the hard ones. The biggest loss is usually not a wrong answer but twelve minutes sunk into one question while two later questions you could have done are never attempted.

So work in two passes. On the first, do every question you can see how to finish; if after about two minutes you have no plan, mark it and move on. On the second, return to the marked questions, most promising first. Keep the last five minutes to check that every answer is actually on the answer sheet.

A question is a time sink when the obvious method is a long case bash or heavy computation and nothing suggests a shortcut. In that situation, look at the options: often four of them fail a quick test.

1.4 Five techniques

1. **Specialise.** If the answer must hold in general, it holds in special cases. Choose values that make the problem trivial.
2. **Test small cases.** For a general n or a large number like 2026, try $n = 1, 2, 3$ and find the pattern.
3. **Work backwards.** If the options are easy to check against the conditions, check them.
4. **Estimate.** If the options are far apart, bounds are quicker than exact work and good enough.
5. **Eliminate.** Parity, divisibility, sign, size and symmetry rule out options without solving the problem.

Example 1.
Which expression equals $1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \cdots + n \cdot n!$ for every positive integer n? (A) $(n+1)! - 1$ (B) $n \cdot n!$ (C) $(n+1)! - n$ (D) $n! + n^2 - 1$ (E) $2 \cdot n! - 1$

Solution.
Try $n = 1$: the sum is 1, and every option gives 1. This is deliberate: setters often make sure the first case rules out nothing. Try $n = 2$: the sum is $1 + 4 = 5$. Now (B) and (C) give 4 and (E) gives 3, while (A) and (D) both give 5. Try $n = 3$: the sum is $5 + 18 = 23$. Option (A) gives $24 - 1 = 23$ and (D) gives $6 + 9 - 1 = 14$. The answer is (A).
Note that we stopped when exactly one option survived, not when the first one fitted. (The proof: $k \cdot k! = (k+1)! - k!$, so the sum telescopes.)

Example 2.
Triangle ABC has $AB = AC = 13$ and $BC = 10$. A point P moves along BC. What is the sum of the perpendicular distances from P to AB and to AC? (A) $\frac{60}{13}$ (B) 9 (C) $\frac{120}{13}$ (D) 10 (E) 12

Solution.
The question gives no position for P and there is no "it depends" option, so the sum must be constant. Put P where the computation is easiest. At $P = B$ the distance to AB is zero, so the sum is just the distance from B to AC, the altitude from B.
The altitude from A is $\sqrt{13^2 - 5^2} = 12$, so the area of the triangle is $\frac{1}{2} \cdot 10 \cdot 12 = 60$, and

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the altitude from B is $\frac{2\cdot 60}{13} = \frac{120}{13}$. The answer is (C). As a check, put P at the midpoint M of BC. Triangle ABM has legs 5 and 12 and hypotenuse 13, so the distance from M to AB is $\frac{5\cdot 12}{13} = \frac{60}{13}$, and by symmetry the sum is $\frac{120}{13}$ again.
The underlying reason is that triangles PAB and PAC together make up ABC, so $\frac{1}{2}\cdot 13\cdot(d_1 + d_2) = 60$ wherever P is. In the exam you do not need it.

Example 3.
Which of the following is closest to

$$S = \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{100}}\,?$$

(A) 10 (B) 14 (C) 18 (D) 22 (E) 26

Solution.
There is no closed form, so do not hunt for one. The options are 4 apart, so any estimate with error below 2 suffices. The crude bound from the largest and smallest terms, $10 < S < 100$, is far too weak.
Instead we bound $\frac{1}{\sqrt{k}}$ between two differences that telescope. Since $\sqrt{k+1} + \sqrt{k} > 2\sqrt{k} > \sqrt{k} + \sqrt{k-1}$, taking reciprocals gives

$$2\left(\sqrt{k+1} - \sqrt{k}\right) = \frac{2}{\sqrt{k+1} + \sqrt{k}} < \frac{1}{\sqrt{k}} < \frac{2}{\sqrt{k} + \sqrt{k-1}} = 2\left(\sqrt{k} - \sqrt{k-1}\right).$$

Sum the left inequality for $k = 1, \ldots, 100$, and the right one for $k = 2, \ldots, 100$, keeping the first term 1 aside:

$$2\left(\sqrt{101} - 1\right) < S < 1 + 2\left(\sqrt{100} - 1\right) = 19.$$

Since $\sqrt{101} > 10$, the left side exceeds 18. So $18 < S < 19$ and the answer is (C).
In general, replace a sum you cannot compute by one you can (here a telescoping sum), and make sure the error is smaller than the gap between options.

Example 4.
The six vertices of a regular hexagon are coloured red, green and blue, each colour on exactly two vertices, so that no two adjacent vertices have the same colour. Colourings that differ by a rotation count as different. How many colourings are there? (A) 12 (B) 16 (C) 20 (D) 24 (E) 28

Solution.
The direct method is a case bash, in which it is easy to miss or double count cases under time pressure. Hence look at the options first.
Take any valid colouring and permute the three colours, for instance by swapping red and green everywhere. The result is still valid, and the six permutations of the colours give six different colourings, because every colour actually appears. So the colourings fall into groups of exactly six, and the answer is a multiple of 6. Only (A) and (D) survive.
It remains to decide whether there are two groups or four, that is, whether there are more than two patterns up to renaming the colours. Write a pattern as a word read round the hexagon from a fixed vertex, naming the colours x, y, z in order of first appearance. The words $xyzxyz$, $xyxzyz$ and $xyzxzy$ are all valid, and different words come from different groups. Three groups already rule out (A), so the answer is (D) 24. There is a fourth pattern,

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$xyzyxz$, which you never needed to find.
This is a common pattern on hard questions: a quick structural fact reduces five options to two, and a small amount of work decides between them.

1.5 Checking without redoing

Repeating a calculation tends to repeat the mistake. Check differently: substitute your answer back, try a special case, or test the size, sign or last digit. On the SMC a wrong answer is 5 marks worse than a right one, so a quick independent check on a doubtful question is worth more than speed.

Common traps.

- Stopping as soon as one option fits a special case. Several options often agree on the first case; keep going until exactly one survives.
- Specialising illegally. If the question says $n > 2$ or that a triangle is acute, your special case must obey that too. A degenerate position, such as a point at a vertex, is fine only if the quantity still makes sense there.
- Guessing on the SMC integer-answer questions. With 1000 possible answers and a penalty for errors, a blind guess is close to a certain loss.
- Guessing blind on the AMC 12. A blank is worth 1.5; a blind guess is worth 1.2 on average.
- Misreading. "Which is not", "closest to" and "rotations count as the same" all change the answer. Underline them.
- Trusting an estimate when the options are close. An estimate settles the question only if its error is smaller than the gap.
- Losing an earned mark at the end: an answer worked out on rough paper but never transferred to the answer sheet.

1.6 How to spot this in a contest

Read the options before you start computing, then run through a short checklist.

- The options are expressions, or the problem says "any" or "every": specialise.
- There is a large index such as 2026, or a general n: compute small cases.
- The options are candidate values that are easy to test: work backwards, most restrictive condition first.
- "Closest to", or options that are far apart: estimate with bounds.
- A long count or computation: look for parity, divisibility or symmetry that rules out options.
- No plan after two minutes: mark it, move on, come back.

Practise all of this on timed past papers. These habits are only useful in the exam if they are automatic before you arrive.

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2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ What is the smallest positive integer that leaves remainder 2 when divided by 3, remainder 3 when divided by 5 and remainder 4 when divided by 7?

(A) 38 (B) 53 (C) 68 (D) 83 (E) 98

Technique: work backwards from the options.

Problem 2 ⋆⋆ $ABCD$ is a square of side 6. A point P is chosen on side CD, with $P \neq D$, and the line AP meets the line BC at Q. What is the value of $\dfrac{1}{AP^2} + \dfrac{1}{AQ^2}$?

(A) $\frac{1}{72}$ (B) $\frac{1}{36}$ (C) $\frac{1}{24}$ (D) $\frac{1}{18}$ (E) $\frac{1}{12}$

Technique: specialise to a convenient position.

Problem 3 ⋆⋆ Which of the following is closest to $\dfrac{1}{1-\cos 1^\circ}$?

(A) 66 (B) 660 (C) 6600 (D) 66 000 (E) 660 000

Technique: estimate.

Problem 4 ⋆⋆⋆ Exactly one of the following numbers is a perfect square. Which one?

(A) 43 693 000 (B) 48 620 001 (C) 52 461 049 (D) 57 320 029 (E) 61 370 026

Technique: eliminate with cheap tests.

Problem 5 ⋆⋆⋆ Distinct real numbers a, b, c satisfy $a+b+c=6$. What is the value of

$$\frac{a^3}{(a-b)(a-c)} + \frac{b^3}{(b-c)(b-a)} + \frac{c^3}{(c-a)(c-b)}\,?$$

(A) 0 (B) 2 (C) 3 (D) 6 (E) 18

Technique: specialise the variables.

Problem 6 ⋆⋆⋆⋆ The numbers $1, 2, 3, 4, 5, 6, 7$ are written in a row in some order $a_1, a_2, \ldots, a_7$. In how many orders is $a_1 + a_2 + a_3 = a_5 + a_6 + a_7$?

(A) 144 (B) 192 (C) 216 (D) 240 (E) 288

Technique: use symmetry to force divisibility, then eliminate.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AMC 12B 2020 Problem 4** ⋆
   Test the options in order, starting from the smallest

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2. **AMC 12A 2015 Problem 6** ⋆
   Work backwards: try each option against the age conditions
3. **UKMT SMC 2026 Question 5** ⋆
   Work out the first few terms
4. **AMC 12A 2016 Problem 6** ⋆
   Estimate the size of N before doing any exact arithmetic
5. **UKMT SMC 2026 Question 15** ⋆⋆
   Specialise: choose a value of n that obeys the stated condition
6. **UKMT SMC 2025 Question 20** ⋆⋆
   Pick convenient x and y satisfying the given relation
7. **AMC 12A 2019 Problem 9** ⋆⋆
   Compute a few terms, spot the pattern, then confirm it
8. **UKMT SMC 2025 Question 21** ⋆⋆
   The answer cannot depend on X, so put X somewhere convenient
9. **UKMT SMC 2025 Question 22** ⋆⋆⋆
   The options let you test a divisibility condition quickly
10. **UKMT SMC 2026 Question 20** ⋆⋆⋆
    Evaluate at a few easy angles to eliminate graphs
11. **AMC 12A 2021 Problem 18** ⋆⋆⋆
    Just evaluate the function on each option
12. **AMC 12A 2020 Problem 9** ⋆⋆⋆
    Sketch both graphs carefully and count crossings

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4 Hints

1. Before testing the conditions in the order given, ask which condition the options were designed to pass. Test the most restrictive one first.
2. The options say the value does not depend on where P is. Is there a position of P where AP and AQ are the same line segment?
3. Convert 1° to radians and use the small-angle approximation for cosine. How accurate does your estimate need to be, given how far apart the options are?
4. All five pass the last-digit test. Try trailing zeros, the digit sum modulo 9, the last two digits modulo 4 and the last three digits modulo 8.
5. The question asserts that the value is the same for every allowed triple. Choose three distinct numbers with sum 6 that keep the denominators small.
6. Whether an order works depends only on the middle number and on which three numbers are on the left. What does that tell you about the answer modulo $3! \times 3!$, and why is the number of suitable choices even? Then look at the parity of a_4.

5 Solutions

Problem 1

What is the smallest positive integer that leaves remainder 2 *when divided by* 3*, remainder* 3 *when divided by* 5 *and remainder* 4 *when divided by* 7*?*

(A) 38 *(B)* 53 *(C)* 68 *(D)* 83 *(E)* 98

Technique: work backwards from the options.

Every option is 8 more than a multiple of 15, so all five leave remainder 2 on division by 3 and remainder 3 on division by 5. Those two conditions eliminate nothing, so the condition modulo 7 decides the answer. Dividing by 7, the remainders are $38 \equiv 3$, $53 \equiv 4$, $68 \equiv 5$, $83 \equiv 6$ and $98 \equiv 0$. Only 53 works, and the smaller option 38 fails, so the answer is (B).

For completeness, by the Chinese remainder theorem the solutions are exactly the numbers congruent to 53 modulo 105, so 53 is the smallest. In the exam you do not need this: one option must be right, and 53 is the only one that passes.

Answer: (B) 53

Problem 2

$ABCD$ is a square of side 6*. A point P is chosen on side CD, with $P \neq D$, and the line AP meets the line BC at Q. What is the value of* $\dfrac{1}{AP^2} + \dfrac{1}{AQ^2}$*?*

(A) $\frac{1}{72}$ *(B)* $\frac{1}{36}$ *(C)* $\frac{1}{24}$ *(D)* $\frac{1}{18}$ *(E)* $\frac{1}{12}$

Technique: specialise to a convenient position.

The value does not depend on P, so choose $P = C$. Then the line AC meets BC at C itself, so $Q = C$ and $AP = AQ = 6\sqrt{2}$. The value is $\frac{1}{72} + \frac{1}{72} = \frac{1}{36}$, answer (B).

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To see why it is constant, let $\theta = \angle DAP$; since P lies on CD and $P \neq D$, we have $0 < \theta \leq 45^\circ$. Triangle ADP has a right angle at D, so $AP = \frac{6}{\cos\theta}$. The line AP makes angle $90^\circ - \theta$ with AB, and triangle ABQ has a right angle at B, so $AQ = \frac{6}{\cos(90^\circ - \theta)} = \frac{6}{\sin\theta}$. Hence

$$\frac{1}{AP^2} + \frac{1}{AQ^2} = \frac{\cos^2\theta + \sin^2\theta}{36} = \frac{1}{36}.$$

Answer: (B) $\frac{1}{36}$

Problem 3

Which of the following is closest to $\dfrac{1}{1-\cos 1^\circ}$*?*

(A) 66 *(B)* 660 *(C)* 6600 *(D)* 66 000 *(E)* 660 000

Technique: estimate.

The options differ by factors of 10, so an estimate that is within a factor of about 3 is enough. In radians $1^\circ = x = \frac{\pi}{180}$, and for small x we have $\cos x \approx 1 - \frac{x^2}{2}$. So

$$\frac{1}{1-\cos x} \approx \frac{2}{x^2} = 2\left(\frac{180}{\pi}\right)^2 \approx 2 \times 57.3^2 \approx 2 \times 3280 = 6560.$$

The answer is (C).

If you want a guarantee rather than an approximation, use $1 - \cos x = 2\sin^2\frac{x}{2}$ together with $\frac{x}{2}\left(1 - \frac{x^2}{24}\right) \leq \sin\frac{x}{2} \leq \frac{x}{2}$. These pin the value between 6565 and 6567, far closer to 6600 than to anything else.

Answer: (C) 6600

Problem 4

Exactly one of the following numbers is a perfect square. Which one?

(A) 43 693 000 *(B)* 48 620 001 *(C)* 52 461 049 *(D)* 57 320 029 *(E)* 61 370 026

Technique: eliminate with cheap tests.

Finding the square root of an eight-digit number by hand is a time sink, and the question tells you that four of the options are not squares. So eliminate them one at a time.

If a square k^2 ends in 0, then 2 and 5 divide k, so $10 \mid k$ and $100 \mid k^2$; dividing by 100 leaves the square $(k/10)^2$. So a square ends in an even number of zeros. Option (A) ends in exactly three zeros, so it is not a square.

A square is congruent to 0, 1, 4 or 7 modulo 9. The digit sum of (B) is 21, so (B) is congruent to 3 modulo 9 and is not a square.

An odd square is congruent to 1 modulo 8, because $(2t+1)^2 = 4t(t+1) + 1$ and $t(t+1)$ is even. A number is congruent modulo 8 to its last three digits, and $029 = 29 \equiv 5 \pmod 8$, so (D) is not a square.

A square is congruent to 0 or 1 modulo 4, and a number is congruent modulo 4 to its last two digits. Since $26 \equiv 2 \pmod 4$, (E) is not a square.

So the answer is (C). Indeed $7243^2 = 52\,461\,049$, but you never needed to find that.

Answer: (C) 52 461 049

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Problem 5

Distinct real numbers a, b, c satisfy $a+b+c=6$. What is the value of

$$\frac{a^3}{(a-b)(a-c)}+\frac{b^3}{(b-c)(b-a)}+\frac{c^3}{(c-a)(c-b)}\,?$$

(A) 0 *(B)* 2 *(C)* 3 *(D)* 6 *(E)* 18

Technique: specialise the variables.

The question asserts that the value is the same for every allowed triple, so pick an easy one. With $(a,b,c)=(1,2,3)$ the three terms are

$$\frac{1}{(-1)(-2)}+\frac{8}{(1)(-1)}+\frac{27}{(2)(1)}=\frac{1}{2}-8+\frac{27}{2}=6,$$

answer (D). A second triple such as $(0,1,5)$ gives $0+\frac{1}{(-4)(1)}+\frac{125}{(5)(4)}=-\frac{1}{4}+\frac{25}{4}=6$ again.

For completeness, here is a proof that the sum S always equals $a+b+c$. Let

$$Q(t)=a^3\frac{(t-b)(t-c)}{(a-b)(a-c)}+b^3\frac{(t-c)(t-a)}{(b-c)(b-a)}+c^3\frac{(t-a)(t-b)}{(c-a)(c-b)}.$$

This is a polynomial of degree at most 2 with $Q(a)=a^3$, $Q(b)=b^3$ and $Q(c)=c^3$, and its coefficient of t^2 is S. So $t^3-Q(t)$ is a monic cubic with roots a, b, c, that is, $t^3-Q(t)=(t-a)(t-b)(t-c)$. Comparing coefficients of t^2 gives $-S=-(a+b+c)$, so $S=a+b+c=6$.

Answer: (D) 6

Problem 6

The numbers $1,2,3,4,5,6,7$ are written in a row in some order $a_1,a_2,\ldots,a_7$. In how many orders is $a_1+a_2+a_3=a_5+a_6+a_7$?

(A) 144 *(B)* 192 *(C)* 216 *(D)* 240 *(E)* 288

Technique: use symmetry to force divisibility, then eliminate.

Whether an order works depends only on the set of three numbers in the first three places and the set in the last three places. Once these two sets are chosen, each can be arranged in 3! ways. Hence the answer is 36 times the number N of ordered pairs of sets (L,R) that work. Swapping L and R turns a pair that works into another pair that works, and $L\neq R$, so these pairs come in twos and N is even. The answer is therefore a multiple of 72.

That eliminates (B) and (D), since 192 and 240 are not multiples of 72. It leaves $144=2\times72$, $216=3\times72$ and $288=4\times72$, so we need the number $M=\frac{N}{2}$ of ways to split the six numbers other than a_4 into two triples with equal sums.

The six numbers sum to $28-a_4$, which must be even, so a_4 is 2, 4 or 6, and each triple sums to 13, 12 or 11 respectively. In each case look at the triple containing 7, which fixes the split. If $a_4=2$, the other two numbers in it sum to 6 and come from $\{1,3,4,5,6\}$, so they are 1 and 5. If $a_4=4$, they sum to 5 and come from $\{1,2,3,5,6\}$, so they are 2 and 3. If $a_4=6$, they sum to 4 and come from $\{1,2,3,4,5\}$, so they are 1 and 3. So each case gives exactly one split, $M=3$, and the answer is (C) 216.

The splits are $\{1,5,7\},\{3,4,6\}$ with $a_4=2$; $\{2,3,7\},\{1,5,6\}$ with $a_4=4$; and $\{1,3,7\},\{2,4,5\}$ with $a_4=6$. Without the elimination step you would still have to find all three, but you would not know in advance that the count had to be a multiple of 72, which is a useful check.

Answer: (C) 216

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The online version, with progress tracking, is at kevinvariant08.github.io/pathways.html.